

ROYAL FITNESS CLUB

Fitness Foundations Certificate

MODULE 2

Exercise Science Fundamentals

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Part of the Royal Fitness Club Professional Certification Program.

SECTION 1: Energy Systems — How the Body Fuels Exercise

The Three Energy Systems

Every movement the human body makes is powered by adenosine triphosphate (ATP) — the universal energy currency of all biological processes. The body cannot store large quantities of ATP; at rest it contains enough for only 2–3 seconds of maximal effort. Three interlinked metabolic pathways continuously regenerate ATP from food substrates.

Understanding which energy system dominates at different exercise intensities and durations is the cornerstone of intelligent programme design — the difference between effective conditioning and random exhaustion.

Pathway 1: Phosphocreatine (ATP-PCr) System

Duration: 0–10 seconds at maximal intensity

The ATP-PCr system (also called the alactic anaerobic system) is the fastest energy pathway. Creatine phosphate (PCr) stored in the muscle donates its phosphate group to ADP, instantly regenerating ATP. No oxygen is required; no lactic acid is produced (hence "alactic").

Enzyme: Creatine kinase (CK) — the rate-limiting enzyme. Activity is highest in fast-twitch fibres.

- Example: 100m sprint start
- Example: Olympic weightlifting (snatch, clean & jerk)
- Example: Shot put, discus throw
- Example: Basketball tip-off jump
- Example: 1-rep max attempt in powerlifting

Recovery: PCr stores recover 50% in ~30 seconds; fully replenished in 3–5 minutes. This governs rest periods in power training — 3+ minutes between true maximal efforts.

Creatine supplementation (3–5g/day monohydrate) increases muscle PCr stores by ~20%, enhancing repeated sprint performance and high-intensity training volume. It is one of the most well-researched and evidence-backed sports supplements.

Pathway 2: Glycolytic System

Duration: 10 seconds – 2 minutes at high intensity

Glycolysis is the metabolic breakdown of glucose (from blood) or glycogen (from muscle stores) into pyruvate. This process regenerates ATP without oxygen — it's faster than oxidative

phosphorylation but less efficient.

Fate of pyruvate: When oxygen delivery is insufficient (high intensity), pyruvate is converted to lactate by lactate dehydrogenase (LDH). This allows glycolysis to continue. Lactate is NOT the cause of muscle fatigue; rather, concurrent hydrogen ion (H⁺) accumulation lowers intracellular pH, inhibiting enzyme activity — the burning sensation during intense exercise.

Step	Substrate	Products	ATP Yield
Fast glycolysis	Glucose/Glycogen	Pyruvate → Lactate	2 ATP (glucose), 3 ATP (glycogen)
Slow glycolysis	Pyruvate	Acetyl-CoA (enters Krebs)	Feeds oxidative system
Lactate shuttle	Lactate	Pyruvate (in liver, heart)	Fuel for other tissues

Lactate threshold (LT): The exercise intensity at which blood lactate begins to accumulate faster than it is cleared. Training near and above LT is highly effective for improving anaerobic capacity. Elite endurance athletes have LT at 85–90% VO₂max; untrained individuals at 50–60%.

Training implication: 200m–800m swimming, 400m–1500m running, 1–3 min HIIT intervals, and high-rep sets (15–25 reps to failure) primarily use this pathway. Rest intervals of 30–90 seconds maintain glycolytic stress.

Pathway 3: Oxidative (Aerobic) System

Duration: >2 minutes (dominates endurance activities)

The aerobic system uses oxygen to completely oxidise carbohydrates, fats, and (to a lesser extent) proteins into CO₂ and H₂O, yielding far more ATP per molecule than glycolysis. It is the primary system for all activities lasting more than a few minutes at submaximal intensity.

Phase	Location	Substrate	ATP Yield per Molecule
Glycolysis	Cytoplasm	Glucose → Pyruvate	2
Pyruvate decarboxylation	Mitochondrial matrix	Pyruvate → Acetyl-CoA	0
Krebs (TCA) cycle	Mitochondrial matrix	Acetyl-CoA → CO ₂	2 GTP
Oxidative phosphorylation	Inner mitochondrial membrane	NADH, FADH ₂ → H ₂ O	~34
Total (from glucose)	—	—	~36–38 ATP
Beta-oxidation (fat)	Mitochondria	Fatty acids → Acetyl-CoA	106–129 ATP per triglyceride

Fat oxidation produces more total ATP but requires more oxygen per ATP molecule. This means at very high intensities (>75–85% VO₂max), carbohydrate becomes the dominant fuel despite fat's higher energy density — the body prioritises speed of ATP regeneration.

Fuel Substrate Continuum

• REST: ~50–60% fat, 40–50% carbohydrate
• LOW INTENSITY (40% VO2max): ~50% fat, 50% carbohydrate
• MODERATE INTENSITY (65% VO2max): ~30% fat, 70% carbohydrate
• HIGH INTENSITY (85%+ VO2max): ~5% fat, 95% carbohydrate
• Post-exercise: Fat oxidation dominates during recovery (EPOC effect)

SECTION 2: VO2 Max — The Gold Standard of Fitness

Understanding VO2 Max

VO2 max (maximal oxygen uptake) is the maximum rate at which the body can consume oxygen during exhaustive exercise. It is measured in ml O₂/kg/min and represents the ceiling of aerobic energy production.

VO2 max is determined by both central factors (cardiac output — how much blood the heart pumps per minute) and peripheral factors (oxygen extraction at the muscle). The Fick Equation captures this relationship:

VO2 max = Cardiac Output max × (a-vO₂) difference max

where (a-vO₂) is the arterio-venous oxygen difference — how much oxygen working muscles extract from arterial blood.

Population	Typical VO2 max (ml/kg/min)
Sedentary adult male	35–40
Sedentary adult female	27–32
Trained recreational male	45–55
Trained recreational female	38–46
Elite marathon runner	70–80
Elite cyclist (e.g., VO2 max record)	~97.5
Thoroughbred racehorse (reference)	~180

How to improve VO2 max: Zone 4 training (85–95% max HR) produces the greatest gains. Classic protocols: 4×4 minutes at 90–95% max HR with 3 min recovery (Norwegian 4×4 protocol). VO2 max improves 15–25% with consistent training in untrained individuals; elite athletes may gain only 2–5%.

Lactate Threshold vs VO2 Max in Programming

While VO2 max sets the ceiling, lactate threshold determines the sustainable percentage of that ceiling. Two athletes with identical VO2 max of 60 ml/kg/min may have dramatically different performance if their thresholds differ:

Athlete	VO2 max	Lactate Threshold	Threshold % of VO2 max	Practical Impact
Athlete A	60	at 75%	75%	Can sustain higher paces before acidosis
Athlete B	60	at 55%	55%	Fades earlier in middle-distance events

This is why experienced coaches focus heavily on threshold training (tempo runs, lactate threshold intervals) alongside VO2 max development — both determine real-world performance.

SECTION 3: EPOC and the Afterburn Effect

Excess Post-Exercise Oxygen Consumption (EPOC)

After intense exercise ceases, oxygen consumption remains elevated above resting levels for minutes to hours. This elevated post-exercise oxygen consumption is called EPOC (Excess Post-Exercise Oxygen Consumption) — colloquially known as the "afterburn effect."

What drives EPOC? Multiple processes require oxygen post-exercise:

- Phosphocreatine resynthesis in muscle (~50% of EPOC)
- Restoration of blood and muscle oxygen stores (myoglobin, haemoglobin)
- Elevated body temperature (every 1°C rise increases metabolic rate ~10–13%)
- Elevated circulating catecholamines (adrenaline, noradrenaline) maintaining metabolism
- Gluconeogenesis — converting lactate back to glucose (Cori cycle in liver)
- Protein synthesis and repair (ongoing after resistance training for 24–48 hours)

Magnitude of EPOC: Low-intensity steady-state cardio produces minimal EPOC (returning to baseline within 30 minutes). High-intensity interval training (HIIT) and heavy resistance training produce EPOC lasting 12–24+ hours. However, total EPOC caloric contribution is often overstated — typically 50–150 extra kcal after a hard 30-min HIIT session.

Practical Application of EPOC

- HIIT burns more total calories than steady-state of equal duration (including EPOC)
- Heavy compound lifts (deadlift, squat, clean) produce greater EPOC than isolation exercises

- Full-body resistance sessions generate more EPOC than split body-part sessions
- Training in a glycogen-depleted state amplifies fat oxidation during and after exercise
- Don't oversell "afterburn" to clients — sustainable consistency matters more than EPOC

SECTION 4: Principles of Progressive Overload

The Foundation of All Training Adaptation

Progressive overload is the most important principle in exercise science. It states that to continually stimulate adaptation, the training stimulus must be progressively increased over time — because the body adapts and what was once a challenge becomes comfortable.

Without progressive overload, adaptation plateaus. With too rapid progression, overuse injury occurs. The art of programming is finding the optimal rate of progression for each individual.

Methods of Applying Progressive Overload

Method	Example	Best Applied To
Increase load (weight)	Add 2.5kg to bench press	Strength/hypertrophy cycles
Increase volume (sets×reps)	Go from 3×8 to 4×8	Hypertrophy blocks
Decrease rest intervals	Rest 90s instead of 120s	Conditioning/metabolic training
Increase frequency	Train muscle 2×/week → 3×/week	Skill, hypertrophy
Improve technique	Deeper squat ROM	Powerlifting, Olympic lifting
Increase density	Same work in less time	Conditioning, fat loss
Increase intensity techniques	Add pause, tempo, bands	Advanced hypertrophy

The 2-for-2 Rule: When an athlete completes 2 extra reps beyond the prescribed rep range on the final set for 2 consecutive sessions, increase the load by the smallest available increment. Simple, effective, evidence-based.

Double progression: First progress reps within a range (e.g., 8–12 reps), then increase weight once the top of the range is reached. Example: 60kg × 8 reps → 60kg × 10 reps → 60kg × 12 reps → 62.5kg × 8 reps. This is the most beginner-friendly model of progression.

SECTION 5: Deload Weeks — Planned Recovery

Why Deloading Is Not Optional

Supercompensation theory describes the cycle of stress → fatigue → recovery → super-compensation (elevated performance baseline). If the next training stimulus arrives during fatigue accumulation rather than super-compensation, fitness declines — a condition called overreaching, or in extreme cases, overtraining syndrome (OTS).

A deload week involves strategically reducing training volume and/or intensity to allow full recovery while maintaining training frequency. It is not rest — it is active management of fatigue.

Deload Type	Volume Reduction	Intensity Reduction	Best For
Volume deload	50–60%	None (same weight)	Hypertrophy athletes
Intensity deload	Same sets/ reps	20–30% weight reduction	Powerlifters peaking
Full deload	50%	20%	Advanced athletes, high fatigue
Active recovery week	Minimal gym work	Very light	Post-competition

Frequency: Beginners may deload every 8–12 weeks; intermediate lifters every 4–8 weeks; advanced athletes every 3–6 weeks, often based on autoregulation (perceived recovery, bar speed, mood, sleep quality) rather than fixed schedule.

SECTION 6: Neural Adaptations — Getting Stronger Without Growing

Strength vs Size — Two Different Outcomes

In the first 4–8 weeks of resistance training, strength gains dramatically outpace visible muscle growth. This is because early strength adaptation is primarily neural — the nervous system learns to operate the muscles more efficiently, not because fibres have grown.

Neural adaptations include:

- Increased motor unit recruitment — more fibres activated per contraction
- Improved rate coding — motor units fire faster (more force per unit time)
- Enhanced motor unit synchronisation — fibres fire more simultaneously
- Reduced antagonist co-contraction — opposing muscles inhibit less
- Improved inter-muscular coordination — muscles work together more efficiently
- Reduced neuromuscular inhibition — Golgi tendon organ becomes less sensitive

These neural adaptations explain why beginners gain strength remarkably fast ("newbie gains") without obvious hypertrophy — and why returning athletes regain strength rapidly after a layoff even before regaining muscle mass.

Key Neuromuscular Concepts

- Motor unit = 1 motor neuron + all muscle fibres it innervates (3–2000 fibres)

• Size principle: motor units recruited small→large (Henneman's size principle)
• Muscle failure occurs when firing rate drops below threshold, not when fibres "run out"
• Eccentric strength is always greater than concentric — exploit this in training
• Rate of force development (RFD) is trainable via explosive and plyometric training

SECTION 7: FITT Principle and Adaptation Types

FITT: The Framework for Programme Design

FITT is the foundational programming framework encompassing four variables that can be manipulated to drive specific adaptations:

Variable	Definition	Example Manipulation
Frequency	How often per week per muscle/modality	Chest: 1×/wk → 2×/wk → 3×/wk
Intensity	How hard (% 1RM, RPE, % HRmax)	Squat: 65% 1RM → 75% → 85%
Time (duration)	How long each session or set	30 min cardio → 45 min → 60 min
Type	Mode of exercise	Running → Cycling → Swimming (cross-training)

Specific Adaptation to Imposed Demand (SAID) Principle

The body adapts specifically to the demands placed on it. Training for strength produces strength adaptation; training for endurance produces endurance adaptation. Programmes must be constructed to match the goal — a common mistake is training for aesthetics with endurance protocols, or training for performance with aesthetic bodybuilding methods.

Training Goal	Rep Range	Sets	% 1RM	Rest	Primary Adaptation
Strength	1–5	3–6	85–100%	3–5 min	Neural, myofibrillar hypertrophy
Power	1–5 (explosive)	3–6	30–70%	3–5 min	Neural, RFD, PCr system
Hypertrophy	6–20	3–5	65–85%	1–3 min	Sarcoplasmic + myofibrillar hypertrophy
Muscular endurance	>20	2–4	<65%	<1 min	Mitochondrial density, lactate tolerance

Module 2 Key Takeaways

- Three energy systems (ATP-PCr, glycolytic, oxidative) work as a continuum — all active simultaneously, with dominance shifting based on intensity and duration

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| • VO2 max sets the ceiling of aerobic performance; lactate threshold determines how much of that ceiling you can sustain |
| • EPOC is real but modest — 50–150 kcal extra after a hard session; training consistency matters far more |
| • Progressive overload is the #1 driver of long-term adaptation — vary the method, not the principle |
| • Deload weeks prevent overreaching and enable supercompensation — they ARE part of training, not time off |
| • Neural adaptations precede hypertrophy — the first 4–8 weeks of gains are largely neurological |
| • SAID principle: train specifically for your goal — strength programmes produce strength, not endurance |

Review Questions

1. What is ATP, and why can't the body simply store large amounts of it?
2. Explain the role of hydrogen ions (not lactate) in muscular fatigue during high-intensity exercise.
3. What is the Fick Equation, and what does each component tell us about VO2 max?
4. Describe the 2-for-2 rule and explain why it is an effective approach to progressive overload.
5. Distinguish between myofibrillar and sarcoplasmic hypertrophy in terms of training stimulus and physiological change.